

1 Understanding Risks to Physical Systems

We conducted a risk assessment to better understand the threat of criminal action against an in-house system. We examined both cybercrimes and physical crimes using data that was normalized to a per-capita rates of crimes. We drew from FBI Internet Crime Center (ICC) annual reports and Chattanooga Police open data records. The population samples were taken from the US Census Bureau reports.

1.1 Introduction

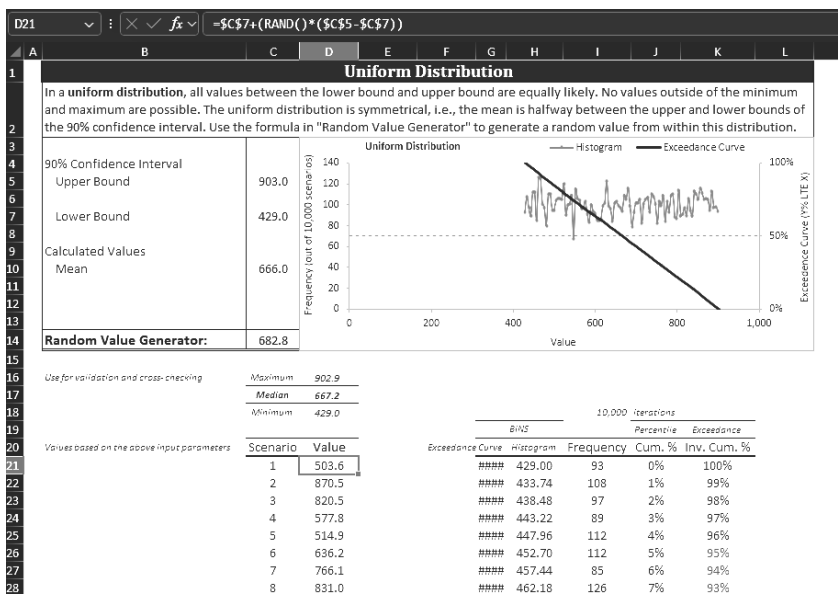


Fig. 1.1: Example run of a Monte Carlo simulation using state-level data to set bounds. Here we chose Personal Data Breach victim counts from 2016 and 2022 as upper and lower bounds. 10,000 rows projected randomized values between these bounds and determined mean and counts by confidence interval. [1]

Background and Content

We had a modest amount of education in Statistics as we began the project. Our skills were derived primarily from three college courses. At Chattanooga State Community College in 2011, we had courses in "Elementary Statistics." [2] This was an algebra-based course that focused on the fundamentals of applying data to normal distributions. Also there in 2012, we had a course in "Statistics for Engineers." [3]

This was an algebra-based course that discussed various quality control techniques and statistical modeling. In 2023 at Harvard University Extension School, we had a graduate course in "Risk in Information Security" with Derek Brink. [4] Professor Brink authored the Excel spreadsheet which we have used for Monte Carlo simulations.

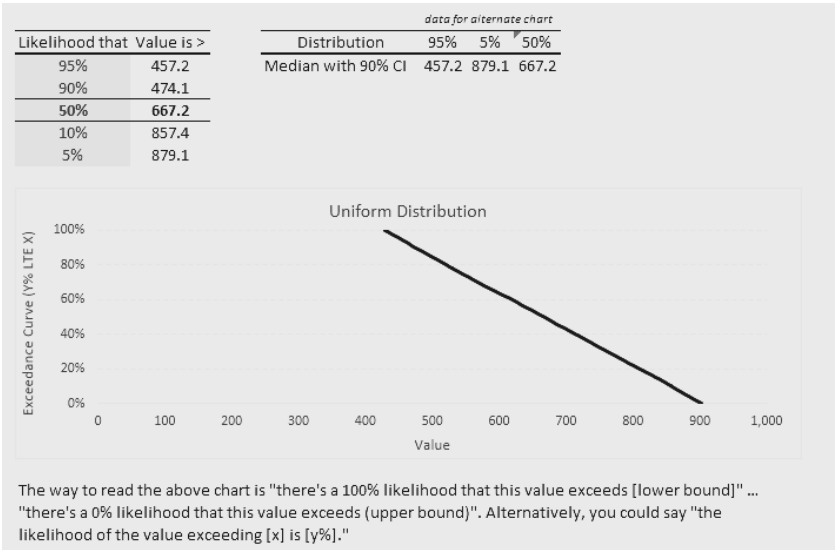


Fig. 1.2: Example results of applying Monte Carlo simulation to forecasting. Here we see the assertion that there is a 90% likelihood that the number of victims will be between 457 and 879; the mean of the simulation was 667.[5]

The spreadsheet conducts 10,000 simulations of data points between user-supplied bounds. Sheets in the Excel workbook are capable of conducting a variety of these simulations. They are differentiated by modeling style (uniform, triangular, normal, binary, or discrete) and points of interest (count, cost, or percentage).

Each of the simulating rows follows these general steps:

- Reference the lower bound.
- Ascertain the range between lower and upper bounds.
- Generate a random number.
- Multiply the random number against the range value.
- Install the generated random value in a cell.

The row values are then counted, sorted, and ranked into confidence intervals. The values and critical points are plotted on various graphs. The end result is that it is easy for an operator of the spreadsheet to generate a Monte Carlo simulation of a distribution between bounds. Since there are enough iterations (10,000), the simulation can have statistical value. Since the inputs on the spreadsheet can be readily modified and since the cells are calculated upon page saving or loading, new simulations can be run with ease. We recognized Professor Brink's spreadsheet was a

useful tool that we could apply to forecasts of interest to us here.

Problem Statement

Project Objectives

Scope of Work

Variables

Assumptions, Risks, and Limitations

We set out to better understand the rates of computer-related crimes in our area using public data sources. We drew from FBI ICC annual reports and Chattanooga Police open data records. The ICC reports provided categorical breakdowns of different types of internet-related crimes. In the past, we were able to find per-state breakdowns of those crimes. The FBI would provide four views of computer crime: overall count, overall financial impact, victim count, and per-victim financial impact. This was available both nationally and per state. When following up in 2026, we noticed we were only able to immediately discover the national-level summaries of crime.

Anecdotally, we know that we have witnessed several of these crimes at rates higher than shown in these tables (Tables 1.2, 1.3, 1.4). We suspect that we're seeing numbers this low because few victims are taking the steps necessary to report computer crimes to federal authorities. For example, when companies or individuals encounter ransomware, their reactions may involve repairing the affected endpoint but not informing law enforcement. Nonetheless, those authorities have been the public's observable source of crime reporting. While we continue to use this data, we may need to accept that true counts of cybercrime in the city must be scaled by an unknown factor.

The Chattanooga Police Department, like other departments of the City of Chattanooga, participate in an open data project. Anonymized crime reports are provided to the public in a manner that lets us track by approximate region of the city and by category of crime. We chose to do this because we recognized some crimes would be reported to local police that were not reported to federal authorities. Some crimes, such as extortion, intimidation, or physical break-ins, were more likely to be handled by local police. Since the systems involved are in-scope for physical damage, we felt the Chattanooga Police Department's open data would be the best public source for supporting criminal threat forecasts for events like burglary from a vehicle. Some policed crimes may focus upon: employees, facility safeguards, loss of computer equipment in public, loss of computer equipment by theft from facilities or vehicles, and so on.

As we examine the data from the public data sources and we feel that the reported rates are less than our experience would justify, we have to recognize that there may be shortcomings in the reported data. Perhaps we would do well to scale the rates up to meet our experience. However, those arbitrary inflations would damage the validity of our academic risk calculations. So, for the time being, we leave

the inflation or deflation of the end results for the reader as a localized adjustment. Since our main goal is to show a method for determining the risks through the reported data, it could be argued that the exact numbers are less relevant than the demonstrated methods.

Significance of the Project

1.2 Literature Review

Overview of Existing Solutions or Approaches

Comparisons of Technologies or Frameworks

Gaps in the Literature

1.3 Methodology

We took cybercrime data from a seven year span of ICC publications. The annual reports for the nation and for each state showed a count of victims per category of cybercrime. The United States (US) Federal Bureau of Investigation (FBI) defined each category for their report. We put that data into a matrix, organized by year and category of crime.

Each state we chose got its own matrix. We noticed that three states typically had high counts of crime; they were: California, Texas, and Florida. We noticed that US territories, like American Samoa, US Virgin Islands, and Puerto Rico, were reported also as if they were a state. These smaller areas with lower populations also showed low counts of cybercrime. We chose these high crime areas (e.g. California) and low crime areas (e.g. American Samoa) as our guides for choosing maximum and minimum limits. Our desired target areas were chosen by the target's home state; our initial runs included Chattanooga, Tennessee.

Using State Data to Forecast a Company's Data

Our main data sets are provided in the public interest; they're based on large units of people like states and nations. As system operators, we have a much smaller interest; we're interested in company sized elements. Instead of being concerned with the cybercrime fate of over 300 million Americans or 9 million Tennesseans, we are probably more interested in what might happen to 100 stakeholders.

To smooth out the counts of cybercrime into rates, we adjusted these state victim counts per capita. We chose the US Census reports for those years to be the population values of the states. This would provide us with a per capita rate of cybercrime based on the chosen state which represented the chosen limit or target in the distribution.

Formally, we note this process below as shown in Figure 1.3. R represents our result. N_t represents our constant for target sample population and is used for multiplying the rate to scale it to match the population of the target. t represents

our starting year for the sample from ICC and coordinates the rows until it progresses to the final year chosen which is denoted by n . i represents our chosen category index of cybercrime. j represents our year index and is built with t until it progresses to the final year of the sample denoted by n . P_j represents the population of the target city at year j , drawn from matrix P .

$$R_j = N_t \sum_{j=t}^{t+6} \frac{X_{j,i}}{P_j}, \quad \text{where } X_{j,i} = \begin{cases} M_{j,i} & \text{if victim count} \\ V_{j,i} & \text{if victim impact} \\ S_{j,i} & \text{if suspect count} \\ D_{j,i} & \text{if suspect impact} \end{cases}$$

Fig. 1.3: Notation for the statistical calculation of the security event matrix adjusted for per-capita rates of incidence.

Five matrices inform this calculation. The first, P as shown in Figure 1.4, records the target city population for each year of the sample as a single column.

$$P = \text{Years} \left\{ \begin{array}{c} t \\ \vdots \\ j \\ \vdots \\ n \end{array} \right\} \left[\begin{array}{c} \text{Population} \\ \hline p_t \\ \vdots \\ p_j \\ \vdots \\ p_n \end{array} \right]$$

Fig. 1.4: Structure of the sample population matrix indexed by year.

The second matrix, M as shown in Figure 1.5, organises the raw cybercrime counts from ICC by category and year, where each entry $a_{j,i}$ is a count that will be adjusted to a per-capita rate.

The third matrix, V as shown in Figure 1.6, organises the victim impact in USD from ICC by category and year, where each entry $a_{j,i}$ is a count that will be adjusted later to a per-capita rate.

The fourth matrix, S as shown in Figure 1.7, organises the count of suspects from ICC by category and year. Each entry $a_{j,i}$ helps us to understand how many suspects are operating within an area of concern. This is a count that will be adjusted later to a per-capita rate.

The fifth matrix, D as shown in Figure 1.8, organises the damages attributed by suspects in USD from ICC by category and year, where each entry $a_{j,i}$ is a count that will be adjusted later to a per-capita rate. This helps us to understand rates of

$$M = \text{Years} \left\{ \begin{array}{c} t \\ \vdots \\ j \\ \vdots \\ n \end{array} \right. \left[\begin{array}{c} \overbrace{\begin{array}{ccccccc} a_{t,1} & \cdots & a_{t,2} & \cdots & a_{t,i} & \cdots & a_{t,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{j,1} & \cdots & a_{j,2} & \cdots & a_{j,i} & \cdots & a_{j,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,2} & \cdots & a_{n,i} & \cdots & a_{n,m} \end{array}}^{\text{Victim Counts by Categories}} \end{array} \right]$$

Fig. 1.5: Structure of the security event matrix for victim counts by cybercrime categories indexed by year.

$$V = \text{Years} \left\{ \begin{array}{c} t \\ \vdots \\ j \\ \vdots \\ n \end{array} \right. \left[\begin{array}{c} \overbrace{\begin{array}{ccccccc} a_{t,1} & \cdots & a_{t,2} & \cdots & a_{t,i} & \cdots & a_{t,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{j,1} & \cdots & a_{j,2} & \cdots & a_{j,i} & \cdots & a_{j,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,2} & \cdots & a_{n,i} & \cdots & a_{n,m} \end{array}}^{\text{Victim Impacts by Categories}} \end{array} \right]$$

Fig. 1.6: Structure of the security event matrix for victim impacts, as in dollars of damage, by cybercrime categories indexed by year.

$$S = \text{Years} \left\{ \begin{array}{c} t \\ \vdots \\ j \\ \vdots \\ n \end{array} \right. \left[\begin{array}{c} \overbrace{\begin{array}{ccccccc} a_{t,1} & \cdots & a_{t,2} & \cdots & a_{t,i} & \cdots & a_{t,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{j,1} & \cdots & a_{j,2} & \cdots & a_{j,i} & \cdots & a_{j,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,2} & \cdots & a_{n,i} & \cdots & a_{n,m} \end{array}}^{\text{Suspect Counts by Categories}} \end{array} \right]$$

Fig. 1.7: Structure of the security event matrix for suspect counts by cybercrime categories indexed by year. These counts consider the number of actors who engage victims in a given area.

effective impact, per suspect, in a given area.

$$D = \text{Years} \left\{ \begin{array}{c} t \\ \vdots \\ j \\ \vdots \\ n \end{array} \right. \left[\begin{array}{ccccccc} & \overbrace{\text{Suspect Damage by Categories}} & & & & & \\ a_{t,1} & \cdots & a_{t,2} & \cdots & a_{t,i} & \cdots & a_{t,m} \\ \vdots & & \vdots & & \vdots & & \vdots \\ a_{j,1} & \cdots & a_{j,2} & \cdots & a_{j,i} & \cdots & a_{j,m} \\ \vdots & & \vdots & & \vdots & & \vdots \\ a_{n,1} & \cdots & a_{n,2} & \cdots & a_{n,i} & \cdots & a_{n,m} \end{array} \right]$$

Fig. 1.8: Structure of the security event matrix for suspect damage, in dollars by perpetrators, by cybercrime categories indexed by year. This tally helps us understand how damaging each actor is to victims.

We apply these matrices to the formula shown in Figure 1.3. The entries $a_{j,i}$ are the raw counts taken from the ICC state-level reports for the chosen source state. Dividing each entry by P_j converts it to a per-capita rate; multiplying by N_t then scales that rate to match the population of the target city. The population values in P provide the year-by-year context for N_t and confirm the representativeness of the scaling. Summing the selected column i of the per-capita-adjusted M across all years yields the final result R .

Table 1.1: Changes to City of Chattanooga’s Population

Year	City of Chattanooga Population	Change Since Previous Year	
2016	177,746	Baseline	[6]
2017	179,530	1,784	[7]
2018	181,918	2,388	[8]
2019	182,799	881	[9]
2020	181,560	-1,239	[10]
2021	181,163	-397	[11]
2022	184,086	2,923	[12]

City population based on the US Census reported population for the City of Chattanooga itself. Surrounding satellite cities and unincorporated areas within the Hamilton County, Tennessee, were not included.

Table 1.2: Cybercrime, City of Chattanooga, Threats to System Intrusion

Year	Data Breach	Personal Data Breach	Malware	Phishing	Ransomware	
2016	1	11	1	7	1	[13]
2017	1	1	1	9	0	[14]
2018	0	18	0	8	0	[15]
2019	0	13	1	8	0	[16]
2020	0	18	0	9	0	[17]
2021	0	19	0	11	1	[18]
2022	1	23	0	7	0	[19]

Counts deduced from IC3 Victim Complaints for Tennessee adjusted for a per-capita rate and then multiplied by the city's population for that year.

Table 1.3: Cybercrime, City of Chattanooga, Threat to Personnel Via Network

Year	Crimes Against Children	Extortion	Threats, Stalking, Harassment, and Terrorism	
2016	0	9	7	[20]
2017	0	7	6	[21]
2018	0	20	7	[22]
2019	0	19	5	[23]
2020	1	31	6	[24]
2021	0	0	4	[25]
2022	1	18	0	[26]

Counts deduced from IC3 Victim Complaints for Tennessee adjusted for a per-capita rate and then multiplied by the city's population for that year. Threats, stalking, and harassment counts were combined with terrorism because the mechanisms were similar.

Table 1.4: Cybercrime, City of Chattanooga, Theft of Data Via Network

Year	Intellectual Property Rights (IPR) and Counterfeit	Identify Theft	Business Email Compromise (BEC)	
2016	1	7	4	[27]
2017	1	8	4	[28]
2018	0	10	7	[29]
2019	1	7	8	[30]
2020	1	10	6	[31]
2021	1	13	8	[32]
2022	0	14	10	[33]

Counts deduced from IC3 Victim Complaints for Tennessee adjusted for a per-capita rate and then multiplied by the city's population for that year.

1.4 Analysis

Using Nation or State Data as a Control for Smaller Forecasts

If we apply unadjusted for sample size counts limited to a one-year basis, we could use a Poisson distribution as shown in Figure 1.9.

$$X_{j,i} = \begin{cases} M_{j,i} & \text{if victim count} \\ V_{j,i} & \text{if victim impact} \\ S_{j,i} & \text{if suspect count} \\ D_{j,i} & \text{if suspect impact} \end{cases}$$

$$Y_i = \sum_{j=t}^{t+6} X_{j,i}$$

$$Y_i \sim \text{Poisson}(\lambda_i E_i)$$

$$R_i = \frac{Y_i}{E_i}$$

Fig. 1.9: Summary of applying Poisson distribution to counts from nation and state data.

This would allow us to feed per-state or national data into the formula and better understand the relationship of our projections to a larger view. It could help us to understand where our projected sample's forecast would stand against the whole state overall or nationally. This way, we could use the state and national data as controls for understanding the context of our forecast.

$$r_{j,i} = \frac{Y_{j,i}}{P_j}$$

Fig. 1.10: Empirical rate per year

$$Y_i = \sum_{j=t}^{t+6} Y_{j,i} \quad E_i = \sum_{j=t}^{t+6} P_j$$

Fig. 1.11: Aggregated count and exposure over historical window

$$\bar{r}_i = \frac{Y_i}{E_i} = \frac{\sum_{j=t}^{t+6} Y_{j,i}}{\sum_{j=t}^{t+6} P_j}$$

Fig. 1.12: Directly standardized (weighted mean) rate

$$Y_i \sim \text{Poisson}(\bar{r}_i \cdot E_i)$$

Fig. 1.13: Poisson model for the aggregated count

$$\bar{r}_i \pm 1.96 \sqrt{\frac{\bar{r}_i}{E_i}}$$

Fig. 1.14: Approximate 95% confidence interval for the rate

$$\hat{R}_i \in \left[\bar{r}_i - 1.96 \sqrt{\frac{\bar{r}_i}{E_i}}, \bar{r}_i + 1.96 \sqrt{\frac{\bar{r}_i}{E_i}} \right]$$

Fig. 1.15: Forecast evaluation: is the forecast within the control band?

1.5 Discussion

Endnotes

- [1]Monte Carlo Modeling Toolkit authored by Derek E. Brink, CISSP, www.linkedin.com/in/derekrbrink. March 2023.
- [2]Triola, Mario F. Elementary Statistics Technology Update + Mystatlab Student Access Code Card. Pearson College Division, 2012.
- [3]Levine, David M, et al. Applied Statistics for Engineers and Scientists. Pearson, 2001.
- [4]Hubbard, Douglas W, and Richard Seiersen. How to Measure Anything in Cybersecurity Risk. Wiley, 2016.
- [5]Monte Carlo Modeling Toolkit authored by Derek E. Brink, CISSP, www.linkedin.com/in/derekrbrink. March 2023.
- [6]<https://www.census.gov/quickfacts/chattanooga-city-tennessee>
- [7]<https://www.census.gov/quickfacts/chattanooga-city-tennessee>
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- [12]<https://www.census.gov/quickfacts/chattanooga-city-tennessee>
- [13]original<https://www.ic3.gov/Media/PDF/AnnualReport/2016State/StateReport.aspx#?s=47> current https://www.ic3.gov/AnnualReport/Reports/2016_IC3Report.pdf
- [14]original<https://www.ic3.gov/Media/PDF/AnnualReport/2017State/StateReport.aspx#?s=47> current: https://www.ic3.gov/AnnualReport/Reports/2017_IC3Report.pdf
- [15]original<https://www.ic3.gov/Media/PDF/AnnualReport/2018State/StateReport.aspx#?s=47> current: https://www.ic3.gov/AnnualReport/Reports/2018_IC3Report.pdf
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- [18]original<https://www.ic3.gov/Media/PDF/AnnualReport/2021State/StateReport.aspx#?s=47> current: https://www.ic3.gov/AnnualReport/Reports/2021_IC3Report.pdf
- [19]original<https://www.ic3.gov/Media/PDF/AnnualReport/2022State/StateReport.aspx#?s=47> current: https://www.ic3.gov/AnnualReport/Reports/2022_IC3Report.pdf
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[26]original:<https://www.ic3.gov/Media/PDF/AnnualReport/2022State/StateReport.aspx?s=47> current: https://www.ic3.gov/AnnualReport/Reports/2022_IC3Report.pdf

[27]original:<https://www.ic3.gov/Media/PDF/AnnualReport/2016State/StateReport.aspx?s=47> current: https://www.ic3.gov/AnnualReport/Reports/2016_IC3Report.pdf

[28]original:<https://www.ic3.gov/Media/PDF/AnnualReport/2017State/StateReport.aspx?s=47> current: https://www.ic3.gov/AnnualReport/Reports/2017_IC3Report.pdf

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[30]original:<https://www.ic3.gov/Media/PDF/AnnualReport/2019State/StateReport.aspx?s=47> current: https://www.ic3.gov/AnnualReport/Reports/2019_IC3Report.pdf

[31]original:<https://www.ic3.gov/Media/PDF/AnnualReport/2020State/StateReport.aspx?s=47> current: https://www.ic3.gov/AnnualReport/Reports/2020_IC3Report.pdf

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[33]original:<https://www.ic3.gov/Media/PDF/AnnualReport/2022State/StateReport.aspx?s=47> current: https://www.ic3.gov/AnnualReport/Reports/2022_IC3Report.pdf